

Arman Hassan

516-373-1256 | armanmansoorhassan@gmail.com | linkedin.com/in/arman-hassan1 | github.com/amansoor

EDUCATION

University of North Carolina at Chapel Hill

Chapel Hill, NC

Bachelor of Science in Computer Science — Minor in Data Science

August 2023 – May 2027

- Coursework: Data Structures, Algorithms, Systems, Computer Organization, Linear Algebra, Models of Computation, Machine Learning, Intro to AI, Mobile Computing, Hardware Security, Data Management

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C, SQL, HTML/CSS

Frameworks: React, React Native, Next.js, Node.js, Expo, FastAPI, Tailwind CSS

Cloud & DevOps: AWS (Bedrock, S3, IAM), Vercel, Docker, Git, GitHub

Databases: PostgreSQL, pgvector, Redis, Amazon S3 Vectors

AI & Data: OpenAI API, Claude, RAG, LLM Agents, scikit-learn, pandas, NumPy, Streamlit

Spoken Languages: English (Native), Urdu (Native), Punjabi (Native), Hindi (Proficient)

RELEVANT EXPERIENCE

Timing

Remote

Software Engineering Intern

January 2026 – May 2026

- Built an **LLM agent (OpenAI API)** that ranks LinkedIn contacts by priority, cutting manual input by **60%**
- Built a **RAG** pipeline with **pgvector** over **10,000+** contacts to generate personalized outreach drafts
- Cut API latency from **~450ms** to **under 90ms** with **PostgreSQL** indexing and **Redis** caching

Introduction to Programming

Chapel Hill, NC

Teaching Assistant

August 2025 – December 2025

- Supported **200+ students** in introductory Python through weekly office hours and lab sessions, debugging code and clarifying core concepts like control flow, functions, OOP, and memory diagrams.
- Graded assignments and collaborated with course staff on feedback to improve student performance.

Vogro (Stanford-affiliated)

Remote

Software Engineering Intern

June 2025 – August 2025

- Built **Python** tools that automated repetitive data tasks, with clear logging and error handling so issues were easy to find and fix, cutting manual processing time by **40%**.
- Created and connected **APIs** using **FastAPI** and **Node.js** that let the **React** website, backend services, and outside tools share data, helping ship features in **3+** releases used by **500+** people.
- Made pages load **20%** faster by reducing unnecessary API calls and cleaning up UI components, so the site ran smoothly for elderly users on older, slower devices.

PROJECTS

Degree Planner AI

Chapel Hill, NC

Personal Project | Next.js, TypeScript, AWS Bedrock, S3 Vectors, Claude, Vercel

September 2026

- Built an AI chatbot on **Vercel** that helps UNC students plan coursework across popular majors and minors
- Built a **RAG** pipeline on **AWS Bedrock** using **Titan** embeddings and **S3 Vectors** to ground **Claude's** answers
- Added an LLM routing layer that identifies each student's program and filters off-topic queries early
- Implemented cost and access controls, including rate limits, an access code, and an automated **AWS** budget cap

Industry Resilience Predictor

Chapel Hill, NC

Carolina Data Challenge Hackathon | Python, pandas, scikit-learn, Streamlit, GitHub, Docker

September 2025

- Built a **machine learning** pipeline to predict how **90+** industries recovered from COVID-era disruptions
- Cleaned and merged **10,000+** time-series data points with **pandas**, handling gaps and misaligned dates
- Trained a **scikit-learn** regression model on this data to estimate each industry's drawdown and recovery trends
- Built an interactive **Streamlit** dashboard to compare industries and visualize predicted recovery metrics
- Used **Docker** and **Git** to keep environments consistent and streamline collaboration across the team

Arcade-Style Space Battle Game

Chapel Hill, NC

Hack110 Hackathon — **1st Place** | Python, Pygame, OOP, Git

April 2024

- Won **1st place** at Hack110 by shipping a full Pygame arcade shooter end-to-end in under 24 hours.
- Built core game systems including collision detection, enemy spawning, and progressive difficulty scaling.